Laboratory work 11

INTRODUCTION TO STEP 7. APLICATIONS

1. Objectives

- Learning the principles of programming on PLC equipment;

2. THEORETICAL BACKGROUND

PROSIM is an advanced process simulatior which works on the base principles of the PLC equipment (Programable Logic Controller – Automate Programabile). All the inputs and outputs are designed in respect of the industrial standards, 24V for digital data and 0-10V for analogic data. The programable automaton – Siemens series S7-1200 – is mounted on a trainer platform which provides the conection with the PROSIM system both with individual and simultan conectors, using a ribbon cable with 50 wires.

Automaton programming, according to international standards, can be done using one of the accepted languages: Ladder Diagram (LAD), Function Block (FUB), Statement List (STL) and Structured Text (ST). The programming environment of the automatons form the S7-1200 series, Step 7 Basic – Totally Integrated Automation Portal, has two of these: LAD and FUB, the user getting to chose. To sustain the safety of the equipments, neither the contacts nor the sensors have a predefined type (normally open or normally closed). Each sensor and each contact are associated to a switch to set the behaviour, letting the user decide on the functioning mode of the contact/sensor depending on the role it has in the equipment and the safety necessities.

Figure 11.1 shows the equipments panel, having three major zones described in the next part.

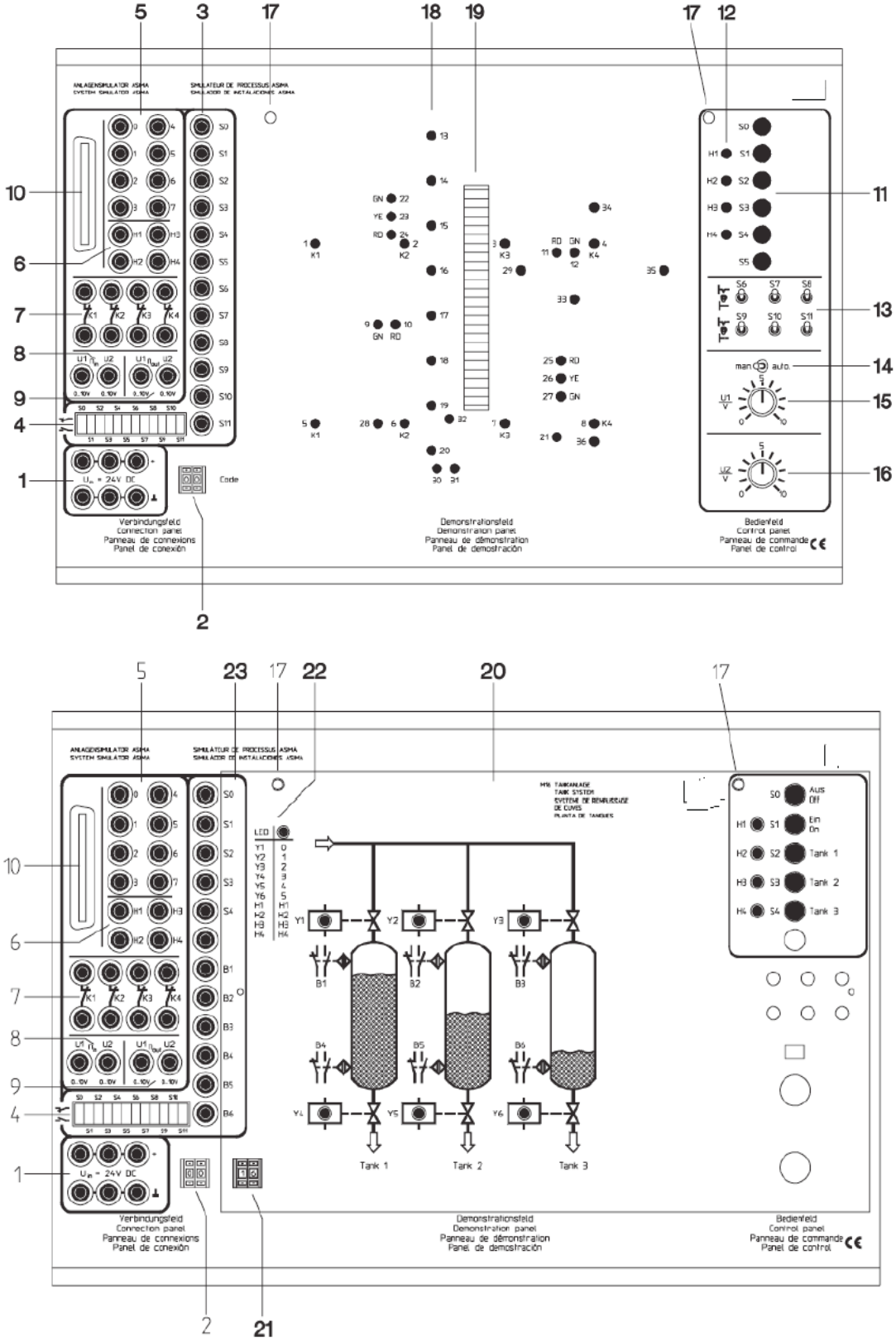

Figure 11.1. The equipments panel descriprion

The connection panel

1. The alimentation terminals of the panel; it needs to be supplied with 24V (the automate supply for example). If the ribbon cable is used, the terminals are not required to supply the equipment

2. Mask selector (it needs to coincide with the number at point 21).

3. Terminals to conncet the sensors and the contacts to the programmable automaton. When using the ribbon cable these terminals are not required.

4. Switch set used in the sensor behaviour setting of the sensors and contacts (normally open or normally closed).

Note: The type of the 12 digital signals (normally open or normally closed, positive or negative logic) which come out of the simulator panel is selected for each part using these switches. For this only the odd position switches are used (the count starting from zero from the left side); and so the second switch from left to right (number 1) will control the behaviour of the signal S0, the fourth (number 3) will control the behaviour of S1 and so on. The switches from even positions are not used and can be on any state without influencing the systems functioning; in this document the switches will be set on the odd neighbours position for simplicity, so, when the afferent switch of S0 is reffered to, the group formed of switch 0 and switch 1 is reffered to, similarly the afferent switch for signal S1 are switch 2 and switch 3 (counting is always done from left to right)

5. Terminals to connect the automaton to the simulator (actuator). When using the ribbon cable, these terminals are not used.

6. Terminals for connceting the automaton to the indicating lights. When using the ribbon cable, these terminals are not used (exception: when the supply is in parallel with another output)

Note: On the masks, the lamps are denoted with the symbol P (P1 for lamp 1, P2 for lamp 2). On the terminals, the afferent signals are denoted with the symbol H (H1,H2). The 2 notations are used for the same signals, in this document we will reffer to them by the P notation.

7. The normally closed contacts of 4 relays are used to ilustrate the safety caracteristics.

8. Analogic inputs in the simulator. The voltage on the terminals respect the industrial 0-10V. When using the ribbon cable, the terinals are not used.
9. Analog outputs from the simulator, they too respect the standard industrial 0-10V. When using the ribbon cable, the terinals are not used.
10. Ribbon cable connector

Control Panel

11. Returning buttons. Their behaviour are set using the switches from group (4)
12. Signaling lights
13. Double function buttons (with or without return). Their behaviour are set using the switches from group (4)
14. „Automatic/Manual" Selector
15. Potentiometer. If the selector (14) is on the "man" position, the set value on the potentiometer (voltage in the interval 0-10V) is applied directly to the process simulator and to the applied mask (for example the rate of which the tank is filled). On the „auto" position of the selector (14) the value set on the potentiometer is ignored, in its place the input U1 is used.
16. Potentiometer. The value set on this potentiometer (voltage in the interval 0-10V) modifies a process parameter (for example the rate of which the tank is emptied)

Demonstrative panel

17. Guide for the mask assigning
18. LEDs which show the state of the processes simulated actuators
19. LED bar which show the movement ot the filling/emptying process of the tanks
20. The applied mask
21. The number of the current mask (The second selector needs to set to this number)
22. Input signal table (lamps and actuators) afferent to the current mask.
23. The output signal label from the simulator (sensors and contacts)

Figure 11.2 presents the programmable automat used (PLC S7-1200 training panel). As it is shown in the equipment presentation, the PROSIM platform is compatible with the industrial ones which use the 24V voltage for its digital inputs and 0-10V for the analogic ones. The interfacing of the signals is made easier, the automate is mounted on an aiding platform, which makes the switch fromthe automates terminals to the terminals and the specific connectors of the PROSIM platform. For example the terminals of the digital inputs Ix.y of the PLC are connected to the digital input panel (figure 11.2 bottom right), the corresponding terminal x.y, and these are connected directly to the connection panel and the asociated terminals (3)– figure 11.1.

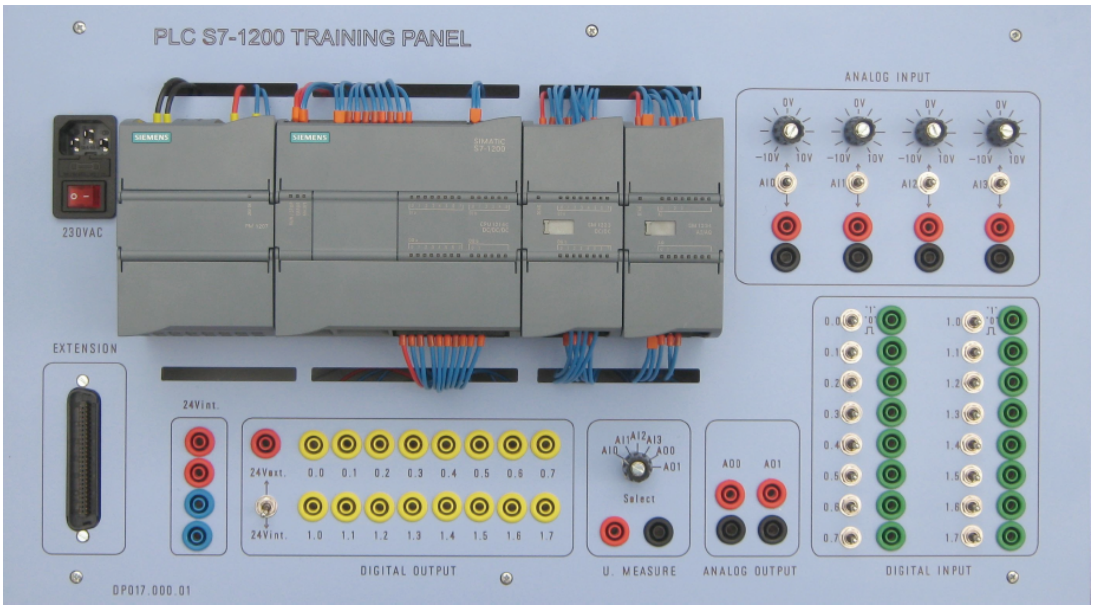

Figure 11.2. Programable automate PLC S7-1200 training panel

4. Problems

The base logic functions, presented in figure 11.3 and presented bellow will be implemented:

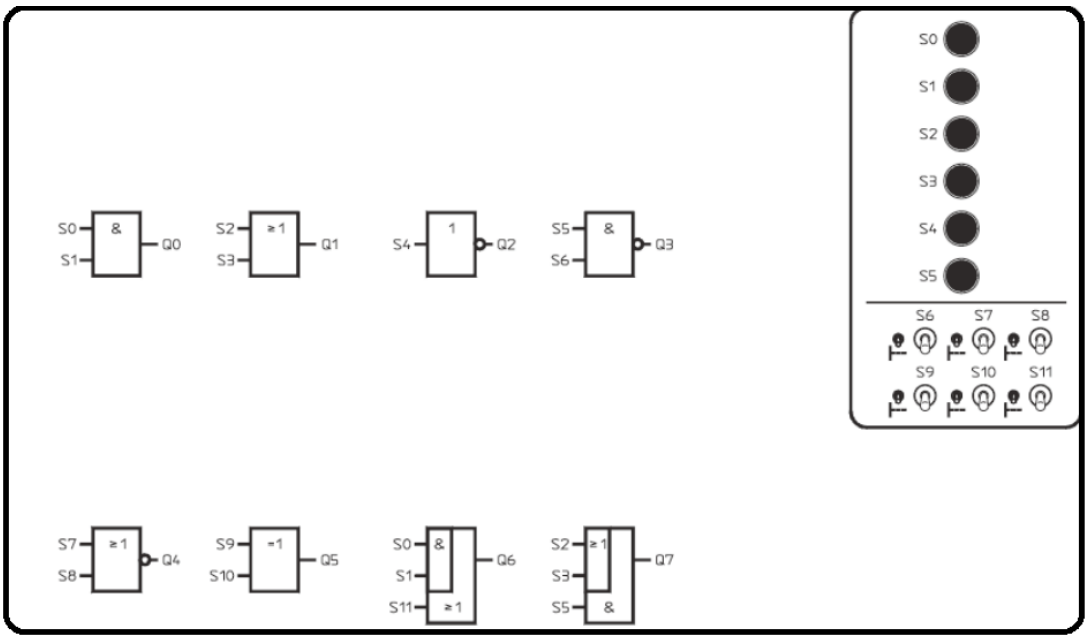

Figure 11.3. The process mask

- AND function: Output Q0 is active if the contacts S0 and S1 are both active

S0	S1	Q0
0	0	0
1	0	0
0	1	0
1	1	1

Q0 = S0 AND S1

- OR function: Output Q1 is active if at least one of the contacts S2 and S3 is active

S0	S1	Q0
0	0	0
1	0	1
0	1	1
1	1	1

Q1 = S2 OR S3

- NOT function (inverter): Output Q2 is ative if S4 is not active

S4	Q2
0	1
1	0

Q2 = NOT(S4)

- NAND function: Output Q3 is active if S5 and S6 are not both simultaneously active

S5	S6	Q3
0	0	1
1	0	1
0	1	1
1	1	0

Q3 = NOT(S5 AND S6)
Q3 = NOT(S5) NOT(AND) NOT(S6)
Q3 = NOT(S5) OR NOT(S6)

- NOR function: Output Q4 is active if S7 and S8 are simultaneously inactive

S7	S8	Q4
0	0	1
1	0	0
0	1	0
1	1	0

Q4 = NOT(S7 OR S8)
Q4 = NOT(S7) NOT(OR) NOT(S8)
Q4 = NOT(S7) AND NOT(S8)

- XOR function: Output Q5 is active if only one of the contacts S9 and S10 is active at a time

S9	S10	Q5
0	0	0
1	0	1
0	1	1
1	1	0

Q5 = (S9 AND NOT(S10))
SAU
(NOT(S9) AND S10)

- AND-OR function: Output Q6 is active if the contacts S0 and S1 are simultaneously active or contact S11 is active

S0	S1	S11	Q6
0	0	0	0
1	0	0	0
0	1	0	0
1	1	0	1
0	0	1	1
1	0	1	1
0	1	1	1
1	1	1	1

Q6 = (S0 AND S1) OR S11

- OR-AND function: Output Q7 is active if at least one of the contacts S2 and S3 is active and contact S12 is active

S2	S3	S12	Q7
0	0	0	0
1	0	0	0

0	1	0	0
1	1	0	0
0	0	1	0
1	0	1	1
0	1	1	1
1	1	1	1

Q7 = (S2 OR S3) AND S12

To implement the logic functions, the application *Totally Integrated Automation* is opened, the option *Create new project* is selected. In the new fields the name of the project (*Project name*) and author (*Author*) are indtroduced. Optional, a short description of the project can be done in the *Comment* section. In the *Path* section the folder in which the project will be saved is selected. After completing the sections the *Create* option is selected and the project will be created.

The next step is to configure the projects principal components, the PLC the digital and analogic input/output blocs. In the new window the option *Configure a device -> Add new device* is chosen. The *CPU 1214C, 6ES7 214-1AE30-0XB0 -> Add* is chosen, just like in figure 11.4 below.

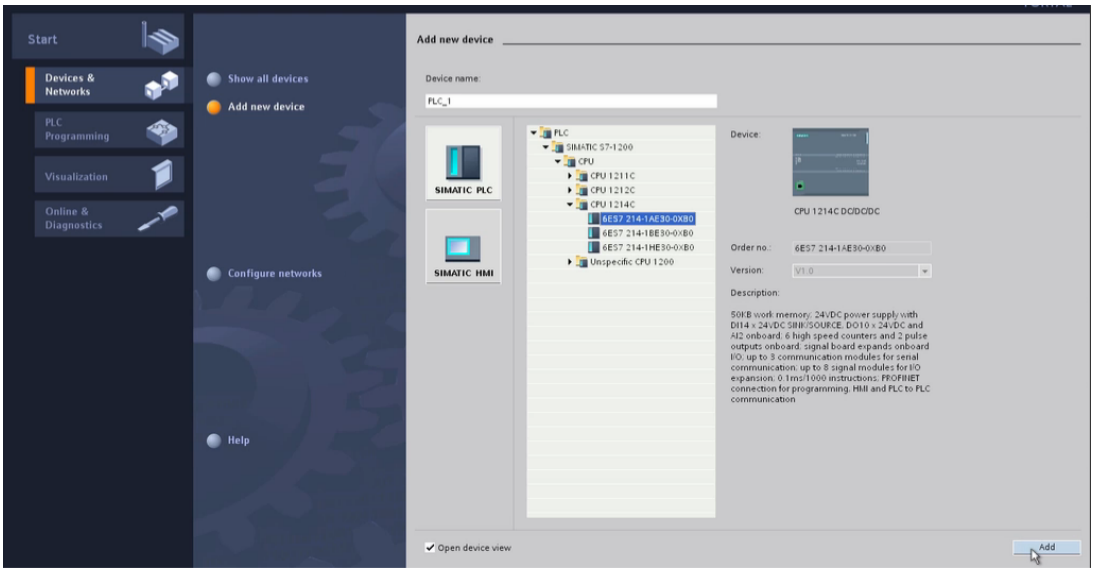

Figura 11.4. Selecting the CPU

In the opened window, figure 11.5, the slots for the principa components can be observed and the added PLC from the previous step.

In the right window we select *Hardware catalog*, following that the digital and analogic input/output modules are added. First *DI/DO ->DI8/DO8 x 24VDC ->6ES7 223-1BH30-0XB0* and added to the project, by dragging and dropping, on the

right of the existing PLC. We select for example, an analogic input, output, *AI/AO->AI4 x 13bits/ AO2 x 14bits -> 6ES7 234- 4HE30- 0XB0* and are added to the project.

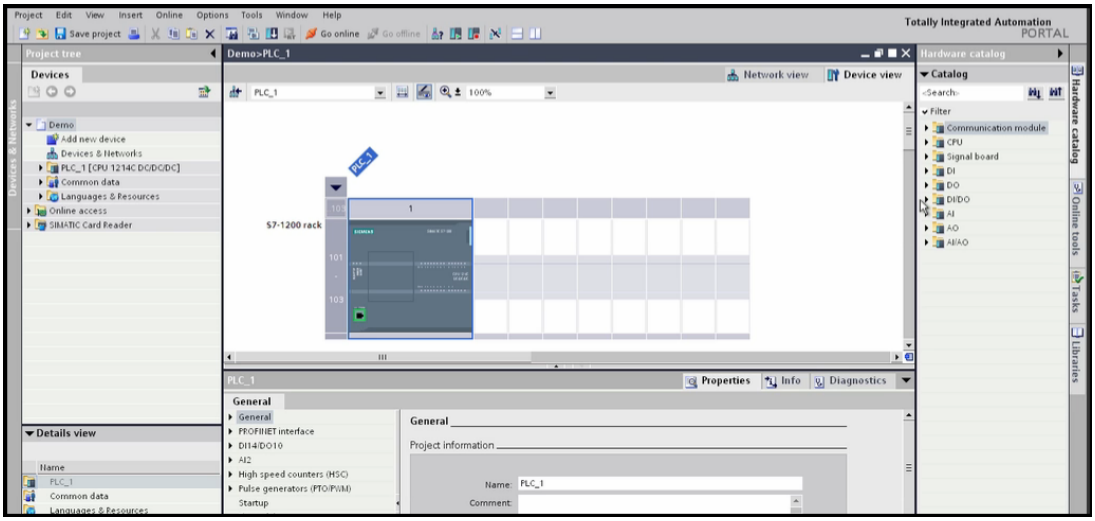

Figure 11.5. Adding the PLC

Following this the PLC is configured and the two added modules. The PLC is selected and in the configuration window we select *Properties -> General -> Project Information -> Name*, introduce *AP S7-1214C*. In the *Author* field the author is specified. After that select *PROFINET Interface -> Ethernet addresses -> Add new subnet*, and specify the necessary TCP/IP to connect the computer which will run the project and the modules (PLC, DI/DO,AI/AO) added to the project. In *IP addresses* inroduce 192.168.0.10, and in *Subnet mask* 255.255.255.0. From *General ->System and clock memory*, check *Enable the use of system memory byte*, and *Enable the use of clock memory byte*. To configure the DI/DO module, select it, in *Properties -> General -> Author*, specify the name of the author. Back at *General ->DI8/DO8 -> IO addresses/ HW identifier* specify the starting address for the input and output: *Input addresses -> Start address -> 2, Output addresses -> Start address -> 2*. Configure the AI/AO module, modify *Properties -> General -> Author*, by introducing the name of the author, and the analogic input starting address (*General ->AI4/AO2 -> IO addresses/ HW identifier -> Input addresses -> Start address -> 3*), and output address (*General ->AI4/AO2 -> IO addresses/ HW identifier -> Output addresses -> Start address -> 3*).

Save the changes: *Project -> Save*.

To write the code select *Project tree*, the name of the previously configured PLC, *AP S7-1214C ->Program blocks ->Main*, which will open the main window of the project. IN the right part of the screen, under *Instructions* the main categories of instruction necessary for the programs which use ladder logic diagrams can be found. Ladder diagrams are special diagrams used in industrial logic control systems. The name ladder comes from their similarity with a ladder: two vertical lines showing the power source, and horizontal lines representing control circuits.

In section *Bit logic*, we can find the instructions necessary for the problems suggested in the laboratory work. Positioning the cursor on any oh=f these instruction give the user information regarding the usage of the instruction. The program is redacted in *Network 1* by including the wanted instructions. For each input instruction type the address corresponding from the DI/DO module is specified, being able to asociate a tag to explain the use of the instruction, figure 11.6.

The output type instructions need to have a reference to the digital output module DI/DO from the programable automate. Similar to the input type instructions it can contain an explication of the use in the tag field, figure 11.7.

The addresses of the input and output signal need to be similar to one of the *digital input*, or *digital output* contacts which are going to be used from the programable automates panel, figure 11.2.

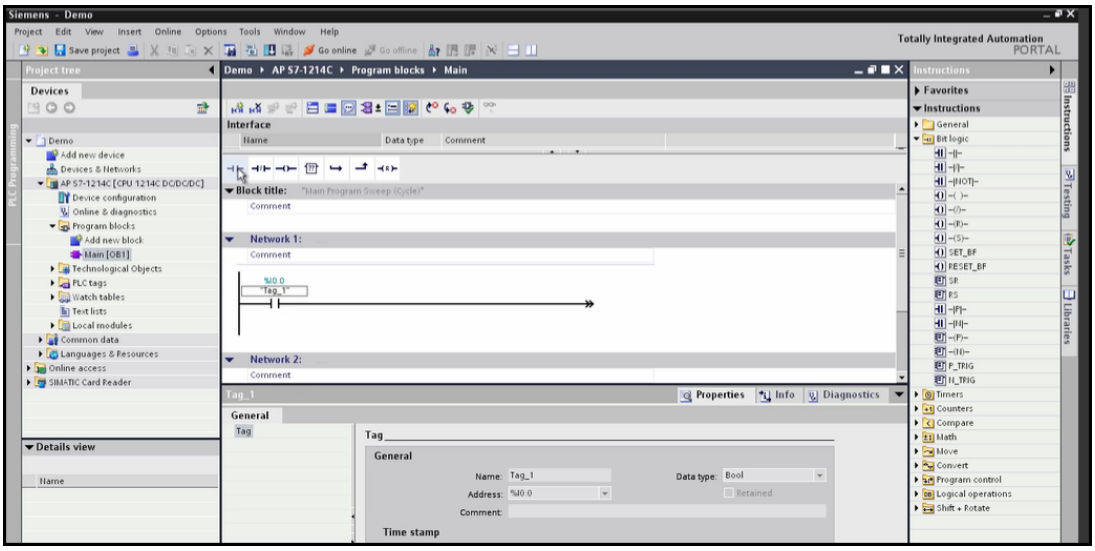

Figure 11.6. Introduction of an input instruction in the main program (main)

To address an input signal the syntagma %Ix.y, where x.y represents the address of the corresponding contact from the programable automate (digital input). For an output signal the syntagma %Qx.y is used, where x.y represents the address of the corresponding contact from the programable automate (digital output).

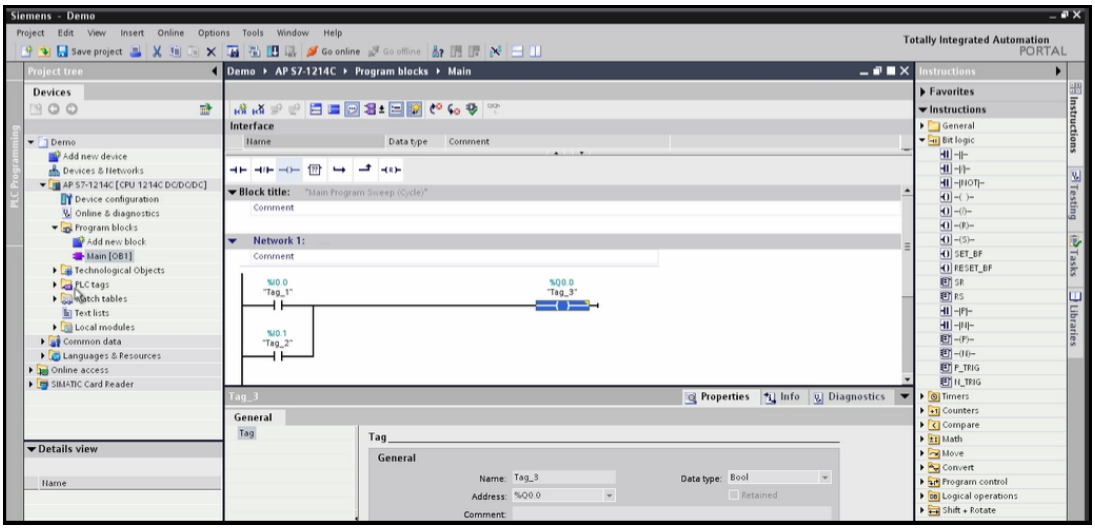

Figura 11.7. Introduction of output instruction in the main program (main)

The most used input instructions are grouped in two categories:

- Opened contact –| |–
- Closed contact –|/|–

Meanwhile the output instruction are in this form –( )–, which can either be open or closed. The principles behind open and closed contaacts is easy to understand. An opened contact will not conduct current until it is activated, while a closed contact will conduct current until it is activated. In ladder diagrams the instructions are executed from top to bottom and from left to right. More instructions in series give the „AND" operator and in parallel give the „OR" operator.

After implementing the logic functions in ladder diagrams and before loading the program on the PLC the following steps need to be done to verify the system configuration:

- Ensure that all the connections are correct
- Ensure that all the Digital Input buttons (PLC S7-1200 TRAINING PANEL) are on the middle position

- Ensure that all the Analog Inputs buttons are on the middle position
- Ensure that the comutator from the Digital Output zone is on the position 24V int (down)
- Ensure that the command panel of PROSIM has all the switches above the potentiometers on the left side
- Use the required mask for the PROSIM platform
- Set the number of the mask on the PROSIM platform (in this case 01)
- Set all the switches (In this case all the switches should be „ON”-up- all the contacts are open)

After verifying the above specified settings the program is loaded on the PLC by selecting *Project tree* the name of the PLC and *Download to device*, as in figure 11.8. In the new window, *Extended download to device*, select the PLC again, and start to load the program by pressing *Load*, which opens *Load preview*. Following that the program is compiled and verified before loading it on the PLC.

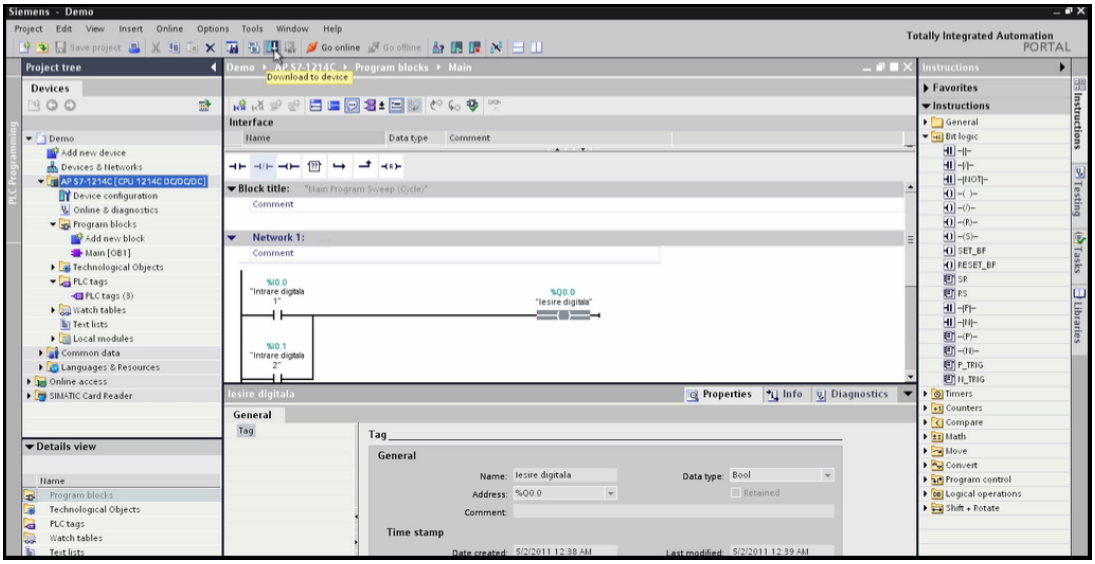

Figure 11.8. Loading the program on the PLC

Successful compilation will be signaled in the *Load preview* window, with the message associated to the PLC: *Ready for loading.* The program is downloaded by checking *Continue* and then *Load*, like in figure 11.9.

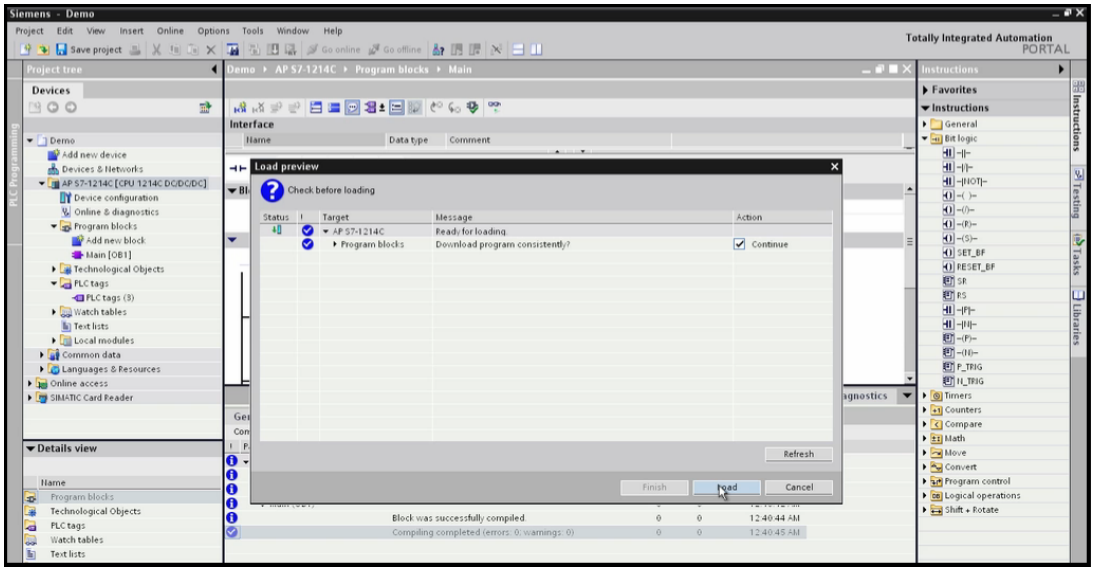

Figura 11.9. Downloading the program on the PLC

In the window *Load results*, all the modules are started by checking *Start all*, followed by *Finish*. All the logic instructions are tested on the PLC after it is downloaded on the device.

Laboratory work 12

STUDY OF THE CONTROLLER PARAMETERS INFLUENCE. PRACTICAL APPLICATION

1. OJECTIVES

- Testing the influence of the PID parameters on the performance characteristics of a closed loop system;
- Exercise the process simulator ACS (Analog Control System).

2. THEORETICAL BACKGROUND

The transfer function of a parallel PID controller is given in the equation:

$$H_R(s) = k_p + \frac{1}{T_i s} + T_d s \tag{12.1}$$

where k_p is the proportional gain, T_i – is the integrators time constant, and T_d- is the derivators time constant. A P controller will reduce the steady state error, but will have a negative effect on the overshoot. An integrating controller - T_i – will remove the steady state error, while a derivative controller – T_d – will increase the stability of the system, by reducing the overshoot and the settling time of the closed loop system.

3. PROBLEMS

Study the influence of the 3 control parameters k_p, T_i and T_d upon the usual performance characteristics of the closed loop system: overshoot, steady state error to a step reference signal and the settling time. The obtained values will be introduced in Table 12.1.

Parameter	Steady state error	Overshoot	Settling time
k_p			
T_i			
T_d			

Tabel 12.1. Controller parameters influence on the performance characteristics

The proposed case study is the control of a DC motor, with the transfer function:

$$H_f(s) = \frac{1}{s(s+2)} \tag{12.2}$$

Using Matlab and different controller types P, PD, PI and PID, study the influence of the parameters on the performance characteristics specified in table 12.1:

$k_p = 5$

$k_p = 10$

$k_p = 5$, $T_d = 0.3$

$k_p = 5$, $T_d = 0.8$

$k_p = 5$, $T_i = 2$

$k_p = 5$, $T_i = 3.5$

Utilizing the process simulator implement the controllers specified before and verify the simulations obtained in MATLAB and the ones obtained in the practical application.

The process simulator, ACS (Analog Control System), contains the necessary blocks to implement the controller system. The blocks ACS – 13001, ACS – 13002, ACS – 13003, ACS – 13004, ACS – 13005, ACS – 13008, ACS – 13010 and ACS – 13016 will be used. Next the blocks will be described.

ACS – 13001 Summing Junction it is used to create the negative feedback of the closed loop system. The input signals of the summing junction are connected on the yellow terminals, meanwhile the output is connected on the blue terminal. Figure 12.1 shows how the summing junction can be used.

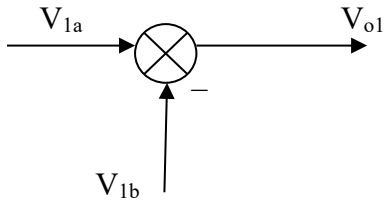

Figura 12.1. Summing junction ACS - 13001

ACS – 13002 P – Controller will ampligy the error signal:

$$\frac{V_0}{V_i} = k_p \tag{12.3}$$

Figure 12.2 shows the principle of the ACS – 13002 block. the value of the proportional gain k_p is set by the white potentiometer (between 0-10), this value can be amplified using the blue potentiometer (1, 10, 50).

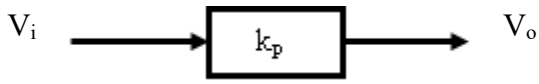

Figura 12.2. P Controller ACS – 13002

ACS – 13003 I – Controller will do the integrating function in a PID controller:

$$\frac{V_0}{V_i} = \frac{k_i}{s} \tag{12.4}$$

The schem of the ACS – 13003 block is in figure 12.3, the value of the parameter $k_i = 1/ T_i$ is set with the white potentiometer (between 0-10), which can be amplified using the blue potentiometer (1, 10, 50).

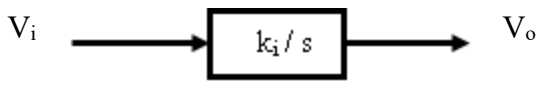

Figura 12.3. I Controller ACS – 13003

ACS – 13004 D- Controller does the derivative effect in the PID controller:

$$\frac{V_0}{V_i} = k_d s \tag{12.5}$$

Schema of ACS – 13004 is given in figure 12.4, the value of the k_d parameter being set with the white potentiometer (between 0-1).

Figura 12.4. D controller ACS – 13004

ACS – 13005 Sum/Dif Amplifier does the sum/subtraction of maximum six signals V_1-V_6, the result being amplified by k, whose value is se using the white potentiometer (between 0-10). Figure 12.5 shows the schema of ACS – 13004 block, with the corresponding equation:

$$V_0 = k(V_1 + V_2 + V_3 - V_4 - V_5 - V_6) \tag{12.6}$$

ACS – 13008 Second Order Plant is used to implement the processes transfer function (12.2).

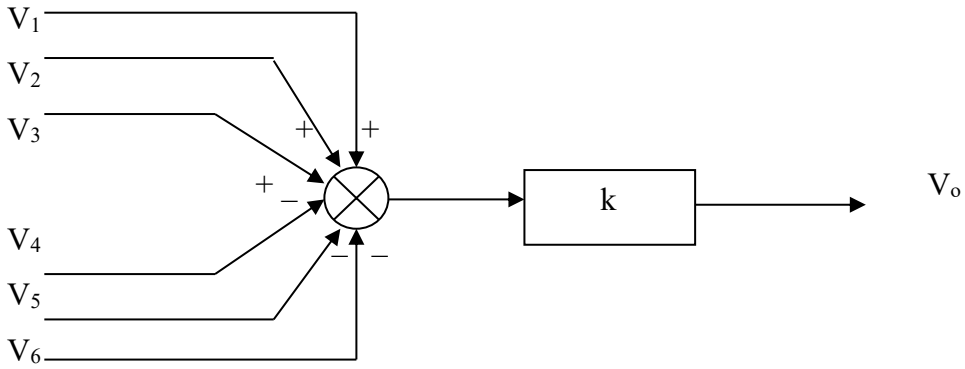

Figura 12.5. ACS – 13005 block

The block lets the user implement both first and second order transfer functions (first order uses an integrator):

$$\frac{V_0}{V_i} = \frac{bT}{s(s+aT)} \quad \text{or} \quad \frac{V_0}{V_i} = \frac{bT}{s+aT} \tag{12.7}$$

The values of the parameters a, b and T can be set by their respective potentiometers: a and b between 0-10, while T with values 1,10 or 100. Figure 12.6 presents the schema of the ACS – 13008 block.

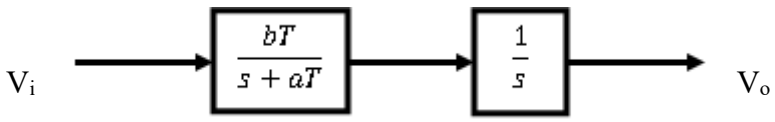

Figura 12.6. ACS – 13008 block

ACS – 13010 Test Signal Generator is used to generate the reference signal in the form of a step signal as shown in figure 12.7. The potentiometers can be used to set the amplitude, offset and frequency.

Figura 12.7. Step signal

Blocul ACS – 13016 Calibration and Testing Module is used to plot the amplitude of different signals: refference, error, control, etc. The output signal of the process can be seen on an external osciloscope connected to one of the terminals CH1 and CH2 of the incorporated adapter of the ACS – 13013 Analog Power Driver module.